

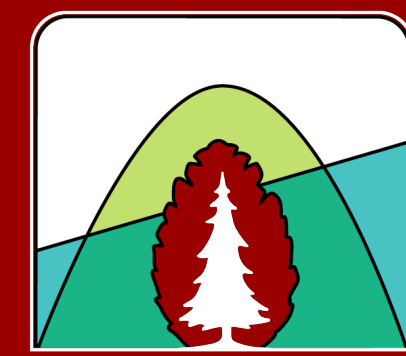


Measuring Emotional Granularity with Music

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PINE LAB



Background

- Emotional granularity describes the degree to which people categorize or differentiate experiences of affect¹
- One approach to measuring emotional granularity involves responding to affective images²
- Music listening is an emotionally evocative experience that allows listeners to traverse emotional space over time and is often cited as a regulatory behavior³
- Listening to music may involve more nuanced experiences of emotion compared to viewing static pictures and thus may be an effective way to measure emotional granularity

Hypothesis: Emotional granularity in response to music and pictures would correlate, but that music would elicit a more granular emotional response than pictures.

Methods

STUDY 1:

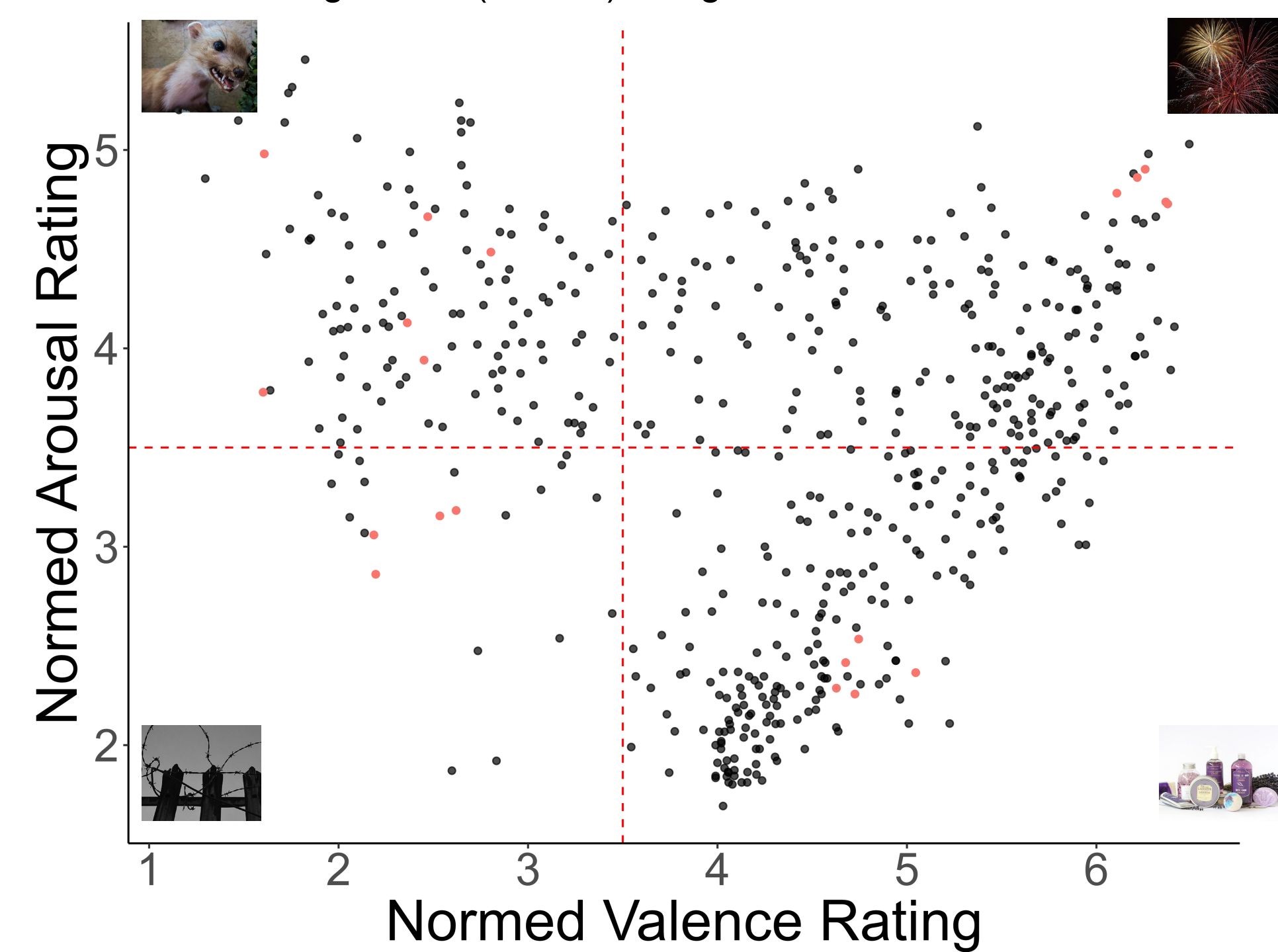
- N = 72 (51F; mean age = 18.6 years)
- Completed emotion rating task (see “Emotion Rating Task”) and psychometric survey battery, consisting of the Goldsmith Musical Sophistication Index⁴ (Gold-MSI), State-Trait Anxiety Inventory⁵ (STAI), Connor-Davidson Resilience Scale-10⁶ (CD.RISC-10), and Beck Depression Inventory⁷ (BDI), and the Range and Differentiation of Emotional Experience Scale⁸ (RDEES)

STUDY 2:

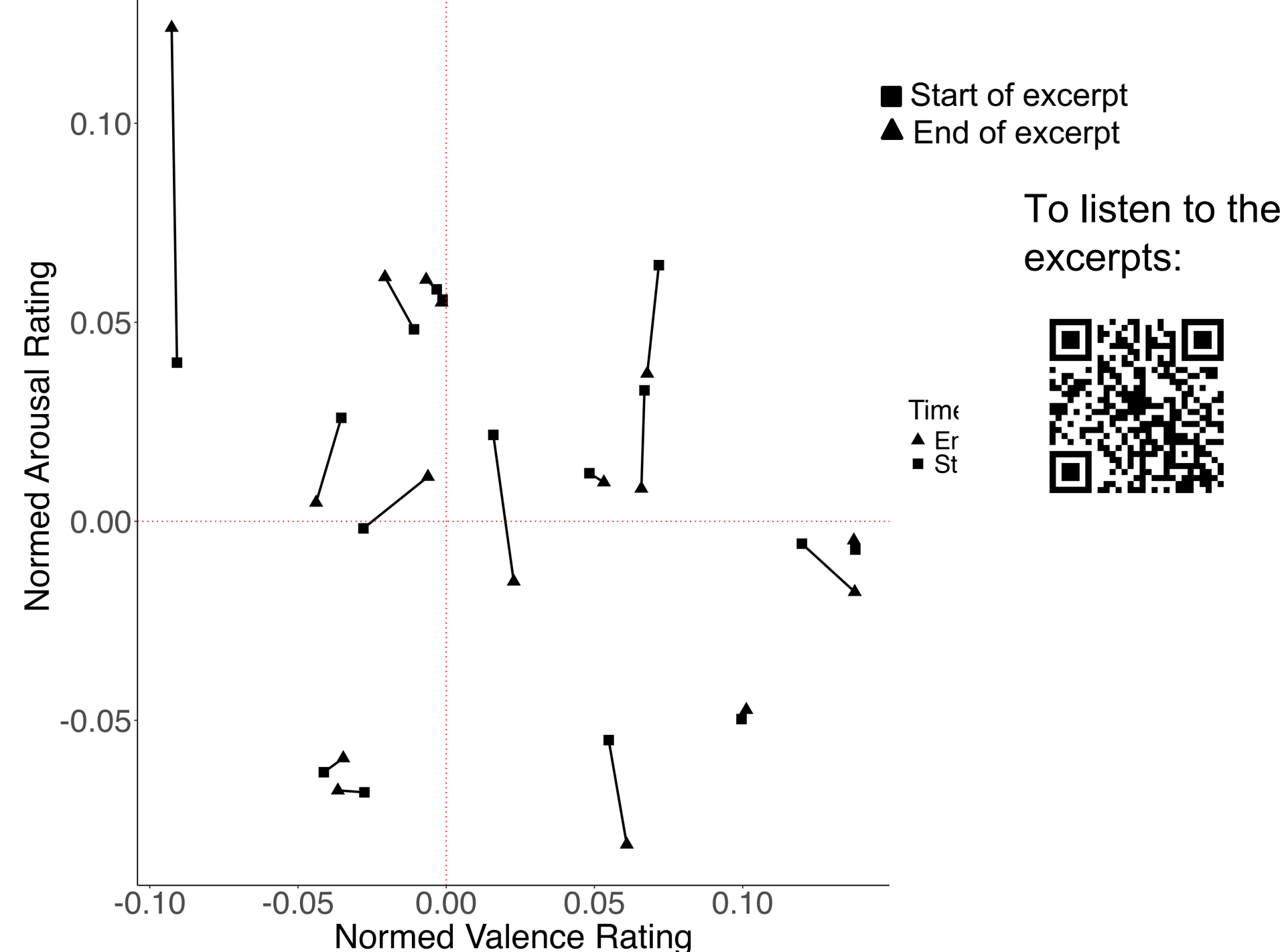
- N = 50 (28F; mean age = 21 years)
- Completed emotion rating task and psychometric survey battery, consisting of Gold-MSI, BDI, RDEES, and Screen for Anxiety Related Disorders (SCAARED)⁹

Stimulus Selection for Emotion Rating Task

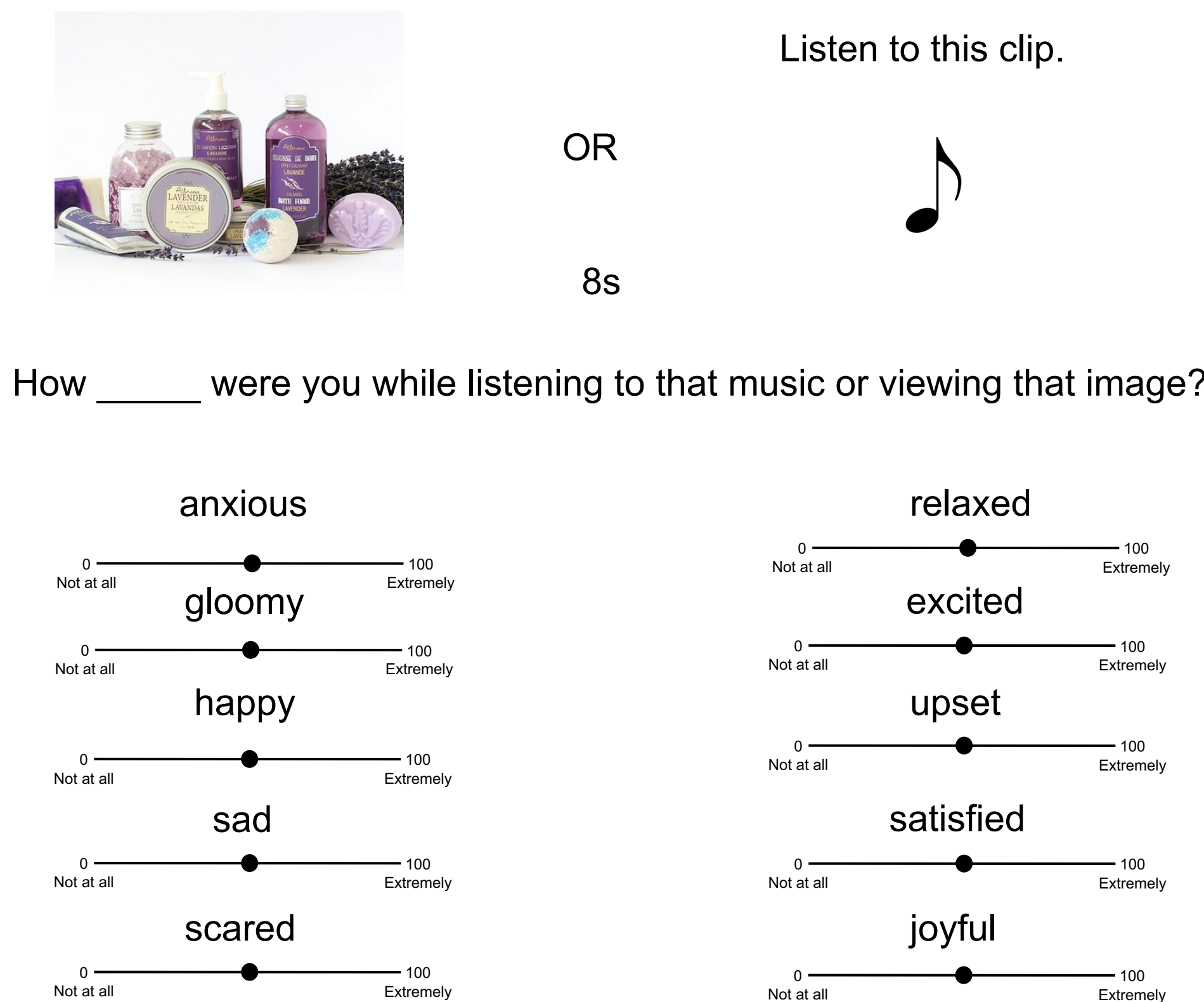
Sixteen affective images (all non-social) drawn from the Open Affective Standardized Image Set¹⁰ (OASIS) using normed data



Sixteen music excerpts drawn from longer music compositions composed to portray specific emotion states¹¹ using normed data

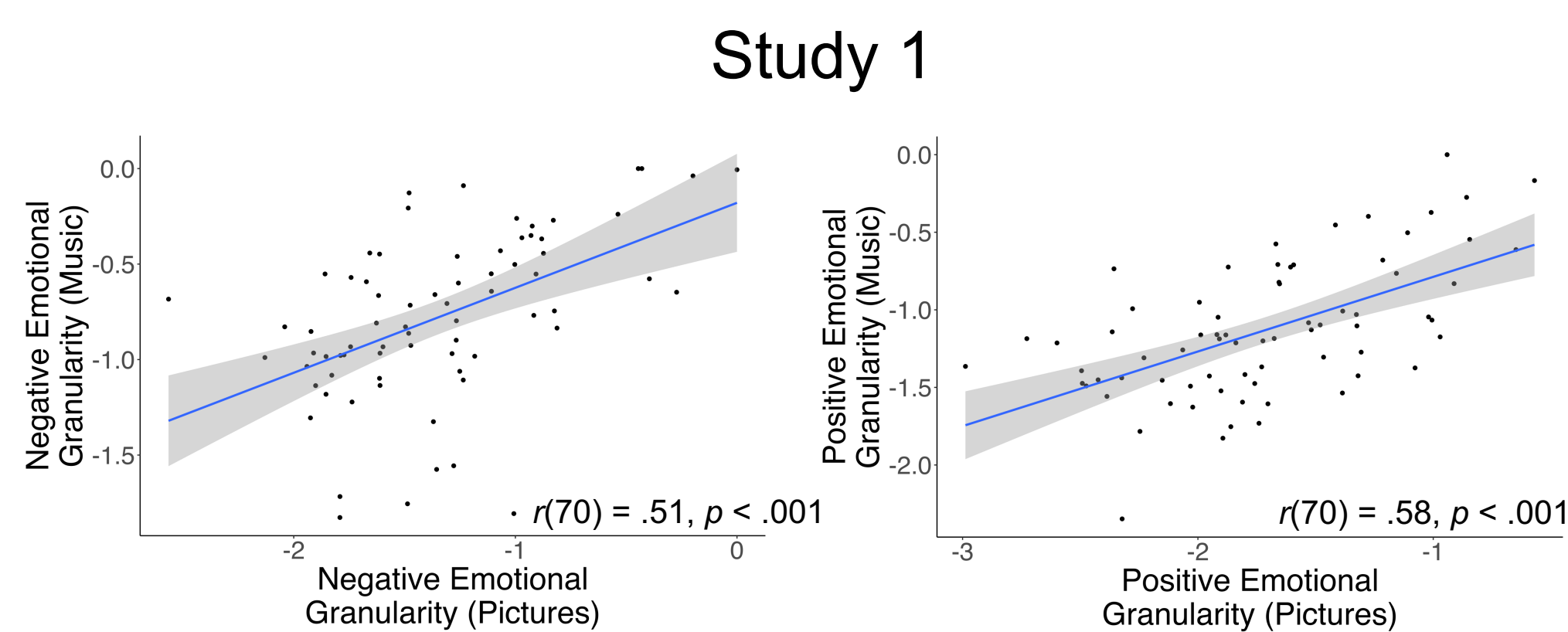


Emotion Rating Task

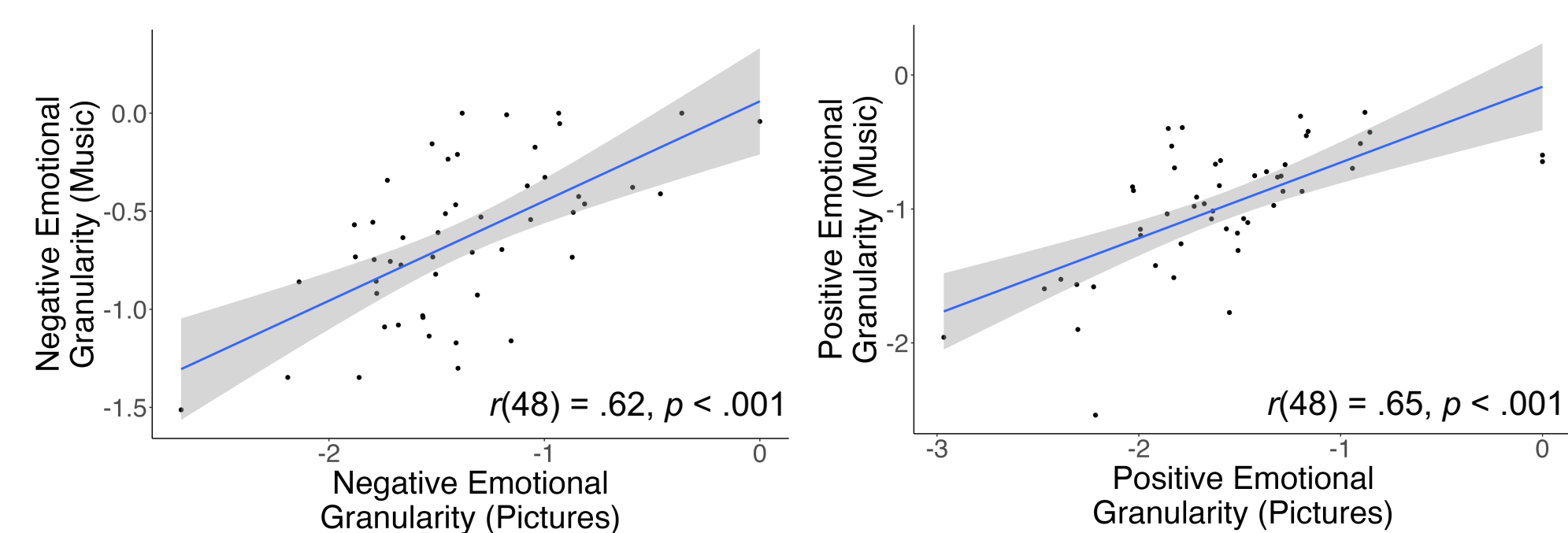


- Emotion words chosen based on those commonly used in emotional granularity tasks¹², along with those commonly cited to be evoked by music¹³
- Emotional granularity scores = -1 X intraclass correlation coefficient (ICC) of emotion ratings (higher scores = greater granularity)¹⁴
- Granularity scores separately calculated for music and pictures and positively- vs. negatively-valenced emotions (4 measures per participant)

Emotional granularity scores are correlated across music and pictures

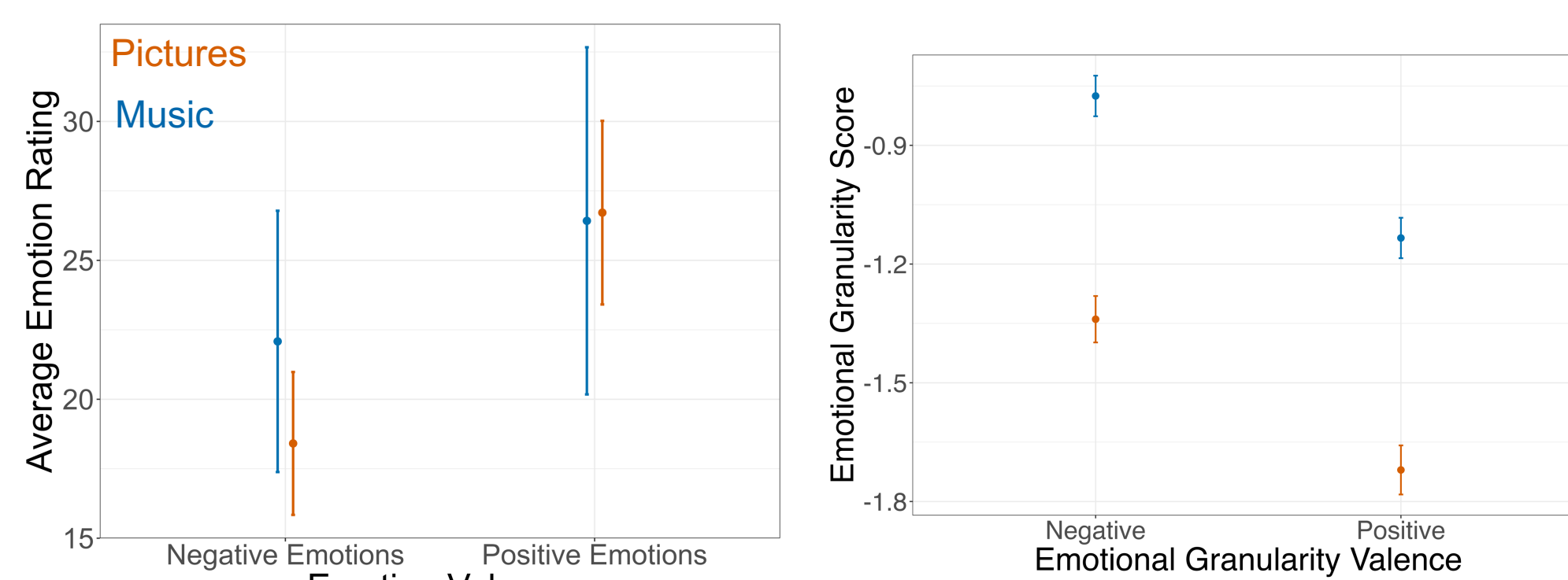


Study 2

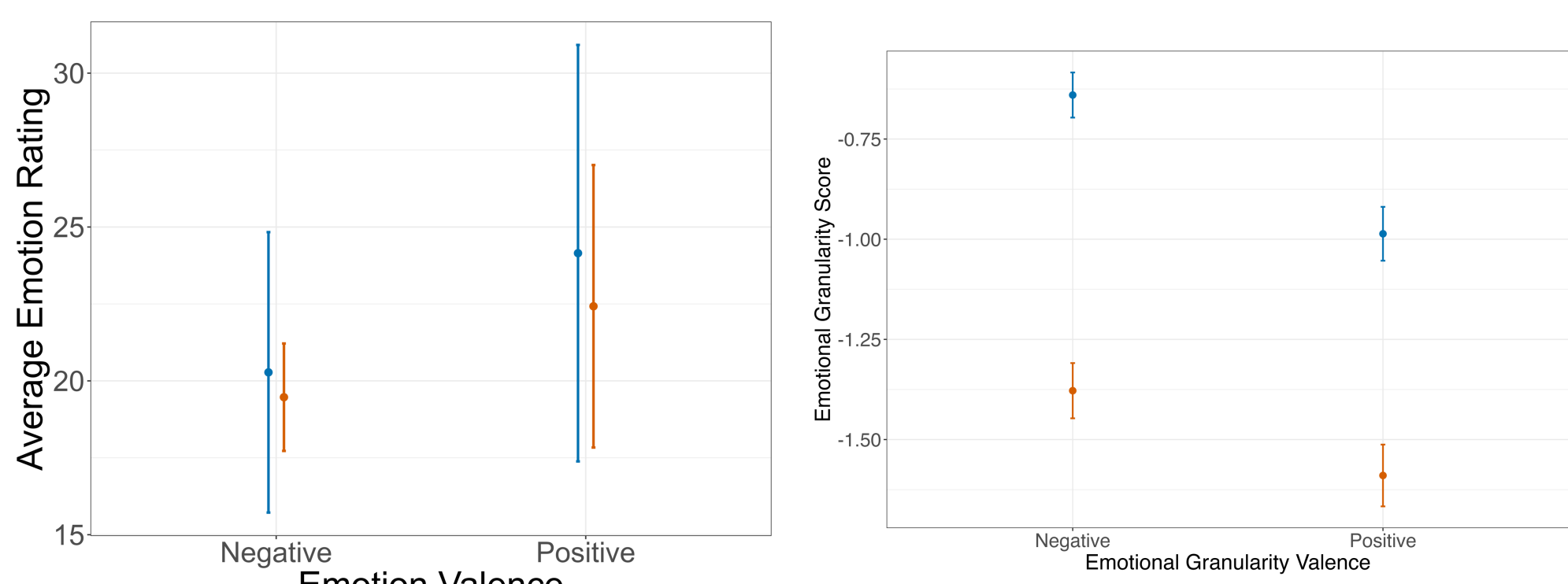


Music and pictures evoked comparable magnitude of emotions, but music elicited more granular emotional responses than images

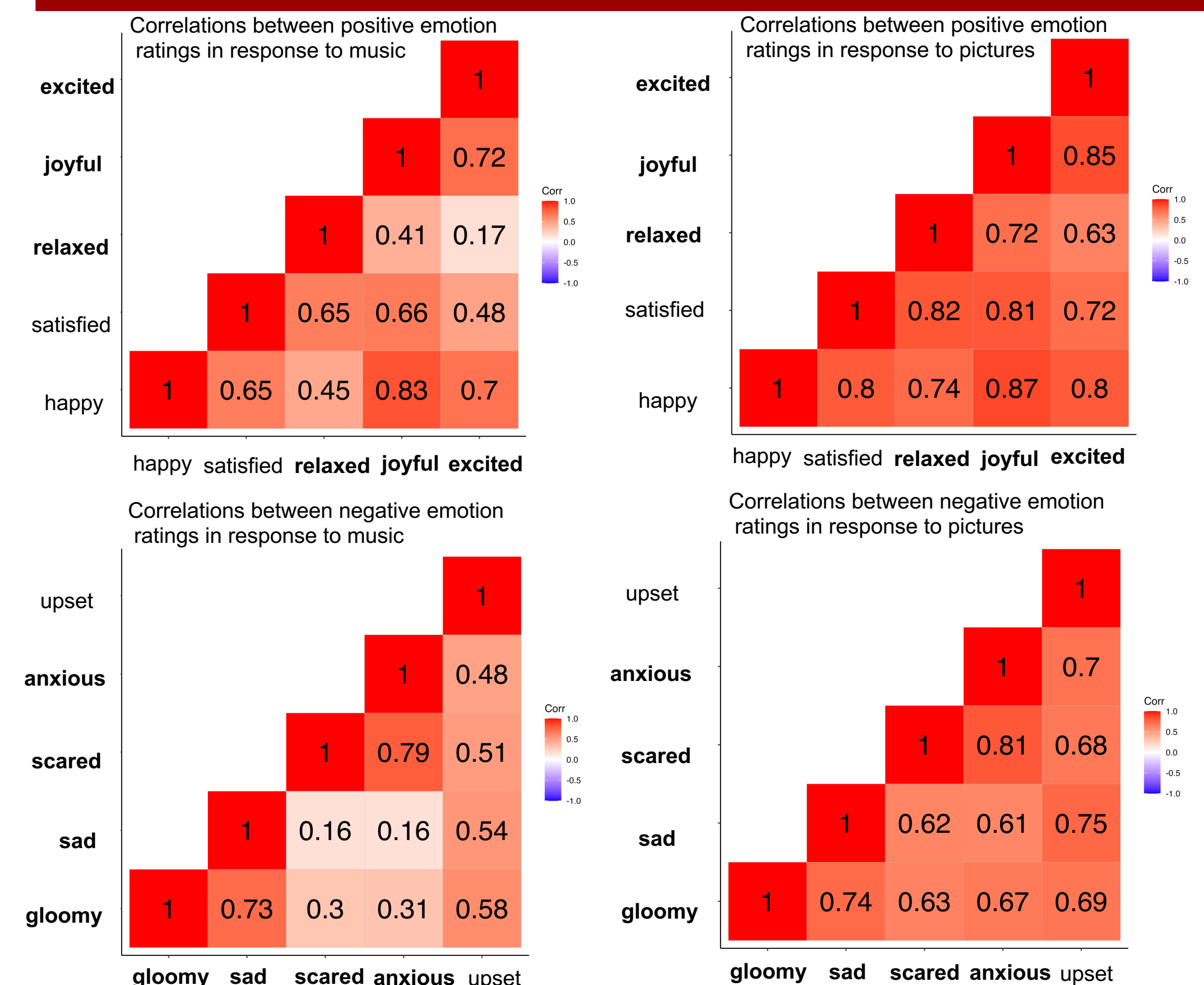
Study 1



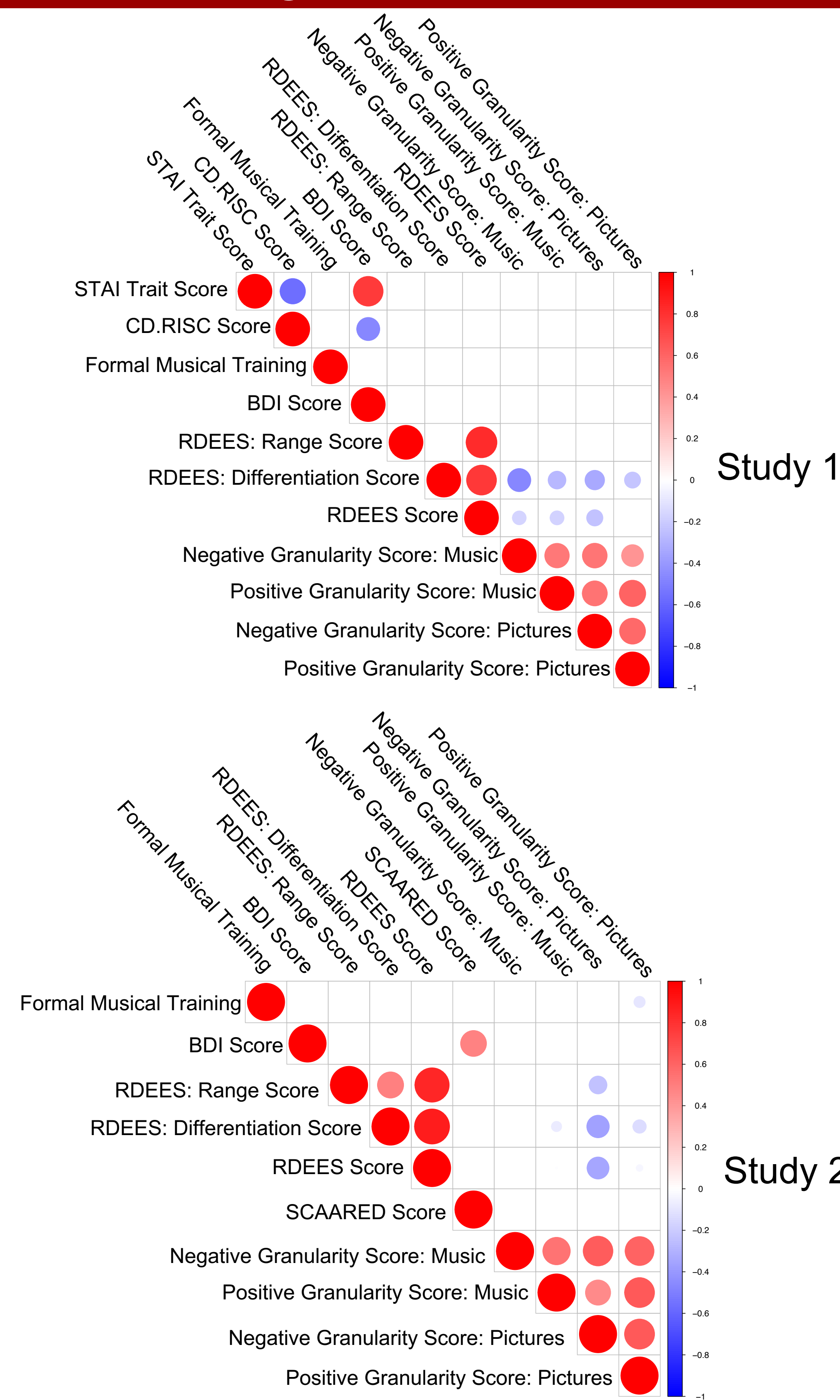
Study 2



Music elicits greater based differentiation for arousal-based images



Emotional granularity experienced in response to music unrelated to musical training and mental health



Study 1

Study 2

Discussion

- Emotional granularity elicited by pictures and music were strongly correlated, suggesting these two approaches are measuring the same psychological construct
- More granular emotional experiences listening to music is potentially driven by greater arousal-based differentiation
- Future studies should disentangle whether greater granularity in response to music is due to greater co-occurrence of emotions or a more dynamic emotional experience

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